Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/28/2010
PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	Risk Assessment of Space Radiation-Enhanced Colon Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7208	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	10/01/2008	**End Date:**	09/30/2011
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: NCE to 9/30/2012 is pending (Ed., 6/22/2011)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Story, Michael (University of Texas Southwestern Medical Center at Dallas) Roig, Andres (University of Texas Southwestern Medical Center at Dallas)		
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Task Description:	This proposal addresses the Carcinogenesis category of the Bioastronauts Roadmap by investigating carcinogenic risk of human colonic cells and tissues after low and high LET radiation exposure. The colon is a major target for the carcinogenic effects of radiation. Aberrant crypt foci are an early precursor lesion for colorectal cancer (CRC) that is characterized by trisomy 7 (present in 37% of adenomas and 80% of adenocarcinomas). The appearance of the adenomatous polyp becomes more prevalent with increased age (10% incidence at age 40 and >25% by age 60). Pre-cancerous lesions may already exist or develop in astronauts going on long-term missions, the result of which would be an increased risk for cancer development through exposure to space radiation. Our key objective is to determine the probabilities for increased colonic carcinogenesis from space radiation as a result of radiation dose, quality, and dose rate. We will expose 2D and 3D human colonic cells and human colonic biopsies to energetic protons (H), high Z and energy particles (HZE), and x-rays and analyze results using both biological assays and genome-wide systems approaches. We will compare these results to a novel immortalized human trisomy 7 colonic epithelial cell line and will model carcinogenic risk of these cells compared to normal and immortalized diploid colonic cells. We will examine DNA damage sensing, signaling, and repair kinetics in normal and trisomy 7 human colonic cell lines. Finally, we will determine if a novel oral bioavailable compound CDDO-Me (bardoxolone methyl, RTA 402) is an effective and non toxic countermeasure for protecting normal colonic cells and tissues from the detrimental effects of low and high LET irradiation. This research should improve our mechanistic understanding for the variability of radiation dose, quality, and dose-rate dependencies for colon cancer development and provide biological markers for modeling and risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of methods to maintain human colonic epithelial cells in culture long term has many benefits for life on Earth. The unique human colonocyte reagents we have developed may help in our understanding of the earliest events in the development of human colon cancer. In addition to normal human colonocytes, we have also isolated the first pre-cancerous human colonocyte cell culture that has one extra chromosome (trisomy 7). Trisomy 7 occurs in a large fraction of patients with sporadic colon cancer and is believed to be one of the earliest steps in colon cancer progression. Our goal is to identify the underlying molecular mechanisms leading to early genetic changes in colon cancer and to validate novel genetic alterations that may occur as critical steps in colon cancer progression. We have initiated experiments to silence and express candidate genes known to be altered in colon cancer into our colonocyte cell lines. Experiments conducting genome-wide sequencing of colon tumors revealed that in addition to the common already known gene mutations, many additional genes are mutated in colon cancer. It is not known which of these mutated genes are important for colon cancer progression. Development of an increasingly cancerous phenotype in the colonocytes after specific genetic alterations should help validate the currently known pathways leading to colon cancer and identify new targets for therapeutic interventions. We are also investigating a novel biological countermeasure that may protect against radiation damage. This would have a major impact on cancer patients being treated with therapeutic radiation.
Task Progress:	1. DNA damage in colonic epithelial cells is more difficult to repair when caused by iron particles compared to protons and gamma rays. (published, see lists at end of this report) 2. Iron damage persists longer in both 2D and 3D colonic epithelial cell cultures compared to protons and gamma rays. (published, see lists at the end of this report) 3. Epithelial cells in intact human colonic tissues display more damage foci than surrounding stromal cells after both high and low LET radiations. All cells in intact colonic tissues are exposed to the same radiation dosage and are likely experiencing equal amounts of double strand breaks, but location in the cell cycle may explain why the replicating epithelial cells show more pronounced DNA damage response compared to quiescent stromal cells. (published, see lists at the end of this report) 4. Additional cells from another patient have been established and characterized. We have an immortalized normal diploid cell line and one that is immortalized with one chromosomal change (trisomy 7). Trisomy 7 is present in about 65% of all colonic adenoma and adenocarcinomas. We have now progressed these cells partially toward tumorigenicity by reducing p53 expression (shRNA to p53), overexpressing K-ras and c- myc and are currently testing if irradiation converts these cells to cancer forming cells. (presented at the St Goar, Satellite meeting, July 2009, manuscript in preparation). 5. We have now demonstrated and published (Gastroenterology, 138:1012-1021, 2010) the ability to differentiate the immortalized human colonic epithelial cell lines We showed that our human colonic epithelial cells express several gastrointestinal putative stem cell markers including CD44, CD29, BMI-1, and LGR5. This indicates that our immortalized human colonic epithelial cells can make similar 3D differentiated structures as do primary cells from mouse small intestine. Our cells are clonal, can divide long-term (are immortalized), and single cells divide to generate self-organizing structures with several differentiated cell types and architecture of crypts present in adults in the absence of a cellular niche. Our cells demonstrate polarized villin staining, junction formation (ZO-1), a subset of cells express Muc2 similarly to the staining of mouse small intestinal cells, and some even express chromogranin A (a neuroendocrine marker). This also helps demonstrate that we are culturing human colonic epithelial stem cells. 6. We have also passaged the cells with trisomy 7 (contain hTERT, cdk4, shRNAp53 and mutant K-ras) and observed that a subclone of these cells contains a trisomy 20 (also very common in colon cancer). While these cells make large colonies and grow anchorage independent in soft agar they do not make tumors in nude mice. We have now expressed c-myc in these cells and these also do not make tumors. These isogenic partially progressed cell lines are extremely useful for radiation experiments to determine biological steps in cancer progression and new experiments are in progress 7. Biological countermeasure experiments: In collaboration with Reata Pharmaceuticals, Inc. (Irving, TX), we have conducted a significant number of experiments with methyl-2-cyano-3,12-dioxooleana-1,9(11)-dien-28-oate (BARD, also termed RTA-402, or CDDO-Me). BARD is an Nrf2 mediated phase II enzyme activator with a well understood molecular mechanism of action. Importantly, BARD is a completely synthetic compound that is non toxic and orally available. In two completed Phase II clinical trials BARD taken orally once-daily not only mimics the body's natural mechanism for resolving inflammation but also reduces the risk of death in patients with stage 3 chronic kidney disease (Reatapharma.com). While anti-oxidant inflammation modulators (AIMs) such as BARD would be expected to provide risk reduction for low-LET radiation, it is much less certain if AIMs would be effective in protecting against high-LET irradiation. In our ongoing research, we have documented that BARD is an effective biological countermeasure by

	reducing high-LET irradiation associated human epithelial cell transformation. BARD protects normal tissues from radiation-associated inflammation through both modulating NF-kB activity, and inducing Nrf2 which acts as a transcriptional activator leading to the suppression of many pro-inflammatory cytokines.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Eskiocak U, Kim SB, Roig AI, Kitten E, Batten K, Cornelius C, Zou YS, Wright WE, Shay JW. "CDDO-Me Protects Against Heavy Ion-Induced Transformation of Human Colonic Epithelial Cells." NASA Human Research Program Investigators' Workship, League City, Texas, February 3-5, 2010. NASA Human Research Program Investigators' Workship, League City, Texas, February 3-5, 2010. Meeting Abstracts. , Feb-2010
Abstracts for Journals and Proceedings	Kim SB, Eskiocak U, Roig AI, Delgado O, Kitten E, Batten K, Wright WE, Shay JW. "Bardoxolone methyl (BARD) is both a countermeasure and mitigator of radiation-induced neoplastic transformation of normal human epithelial cells." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 17-19, 2010. 21st Annual NASA Space Radiation Investigators' Workshop, May 17-19, 2010. Meeting Abstracts. , May-2010
Articles in Other Journals or Periodicals	Roig A, Shay JW. "Immortalization of adult human colonic epithelial cells extracted from normal tissues obtained via colonoscopy." Nature Protocols Network 2010. http://dx.doi.org/10.1038/nprot.2010.29 , Jul-2010
Articles in Peer-reviewed Journals	Roig AI, Wright WE, Shay JW. "Is telomerase a novel target for metastatic colon cancer?" Current Colorectal Cancer Reports. 2009 Oct;5(4):203-8. http://www.citeulike.org/article/5956107 , Oct-2009
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Articles in Peer-reviewed Journals	Roig AI, Eskiocak U, Hight SK, Kim SB, Delgado O, Souza RF, Spechler SJ, Wright WE, Shay JW. "Immortalized epithelial cells derived from human colon biopsies express stem cell markers and differentiate in vitro." Gastroenterology. 2010 Mar;138(3):1012-21.e1-5. Epub 2009 Dec 4. PubMed PMID: 19962984 , Mar-2010
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